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VisualCS: Enhancing Visualization in Computer Science Education Through AI-Powered Visual Generation

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WHY VISUALS MATTER

CS is abstract. Visuals make it tangible.

Algorithms, data structures, and systems can't be directly observed – visualization is how students build a mental model of what the code is doing.

TEACHERS AGREE

100%

of CS educators surveyed agreed that visualization helps students learn.

STUDENTS STAY IN CS

88% vs 47%

Retention for intro course in CS students using visualization-based tools, vs. peers without.

Sources: Naps et al., 2002; Fouh, Akbar & Shaffer, 2012

THE REALITY

Building good visuals is hard.

Prior research identifies three barriers that keep instructors from using visualizations even when they want to.

01

Domain knowledge

Clear, accurate diagrams require deep understanding of CS concepts. Not every topic is easy to draw.

02

Visual design skills

Good diagrams take layout, hierarchy, and clarity. Skills most instructors were never trained in.

03

Tool learning curve

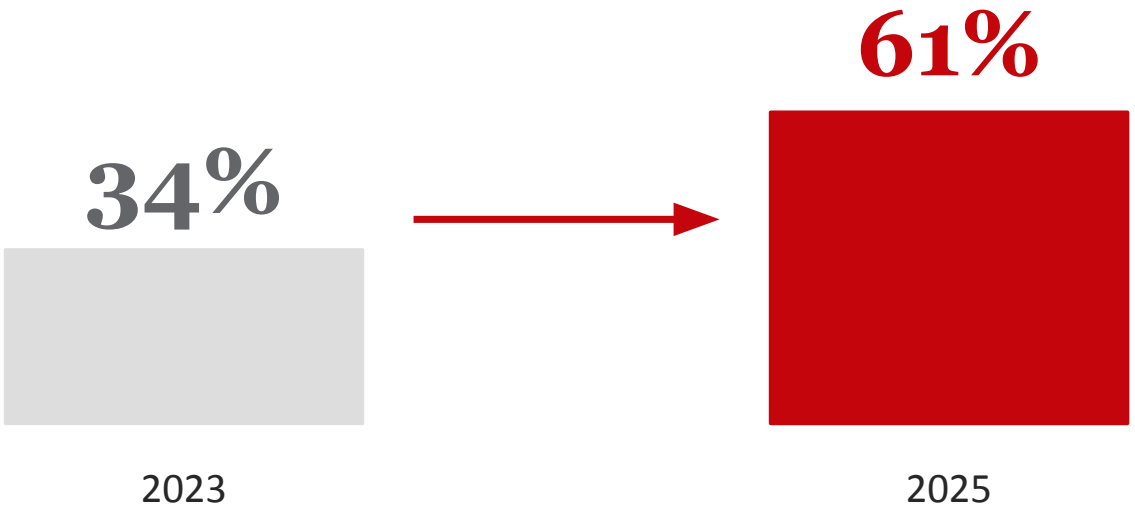
Specialized drawing software takes time to learn, and time to use every single lecture.

Sources: Naps et al., 2002; Shaffer et al., 2011

WHY NOW

AI is here, but not for visuals.

Teacher adoption of AI nearly doubled in two years.



But mostly for text.

Lesson plans, summaries, grading, writing feedback, **not** visual content.

Of 71 peer-reviewed studies on AI in CS classrooms, only 1 was about visual content.

Sources: EdWeek Research Center, 2025; Prather et al., 2024

RESEARCH QUESTION

So we ask:

RQ1 What challenges do CS instructors face when creating visualizations for teaching in the age of AI?

RQ2 How can an AI-assisted tool address those challenges?

FORMATIVE STUDY

What instructors actually need

10 semi-structured interviews with CS Professor and Teaching Faculty

01

Trust in images

AI images often have garbled text, wrong numbers, and unrealistic details. Instructors also worry students will see AI visuals as low-effort.

02

Accessibility first

Instructors avoid color because to accommodate color blindness. Visuals also need alt text. Simple box and triangle work best.

03

CS Unfolds

Some concepts fit a single diagram. Others algorithms, data flow, state changes need sequential steps to make sense.

04

Control matters

Instructors want to customize colors, edit text, and build consistent visual languages across diagrams.

Based on reflexive thematic analysis of 10 semi-structured interviews.

Introducing VisualCS

A web tool that lets CS instructors generate custom diagrams from plain descriptions.

CS concept to visualize *

e.g. binary search tree insertion

Class name

e.g. Algorithms

Student level

Intro

Example data (optional)

e.g. use array [3, 7, 12, 18, 24, 31, 45], searching for 18

Special instructions / teaching style (optional)

e.g. always show index numbers below array cells, use pseudocode notation from CLRS, avoid recursion diagrams

☐ Generate alt text ☐ Advanced mode (review scene description)

Generate visualization

- **Highly Customizable**
  Instructors provide their course level, teaching style, and custom prompt, so visuals fit their specific classroom.
- **Two-step AI pipeline**
  The text model plans the diagram first, then the image model renders it so the content stays correct.
- **Step-by-step breakdown**
  AI decides when a concept needs multiple steps. Structures image get 1, while algorithms get more. More steps mean multiple images generated to show the flow.

Highly Customizable

01

Student level

Sets the level of detail

Simpler visuals with fewer steps for beginners. More detail and more steps for advanced learners.

02

Class name

Controls style

Different courses draw the same idea differently. The tool picks the right style for each one.

03

Special instructions

Overrides everything

Always show index numbers or Use textbook notation or Highlight changes in red. Whatever the instructor asks for, applied across every visual.

Why Customizable?

Instructors shouldn't have to become prompt engineers to use AI visuals. Just set once for learner level, course style, personal rules and VisualCS builds the right prompt every time.

DEMO

Example

CS concept to visualize *

Binary search tree

Class name

CS 300, Intro to Computer Science

Student level

Intro

Example data (optional)

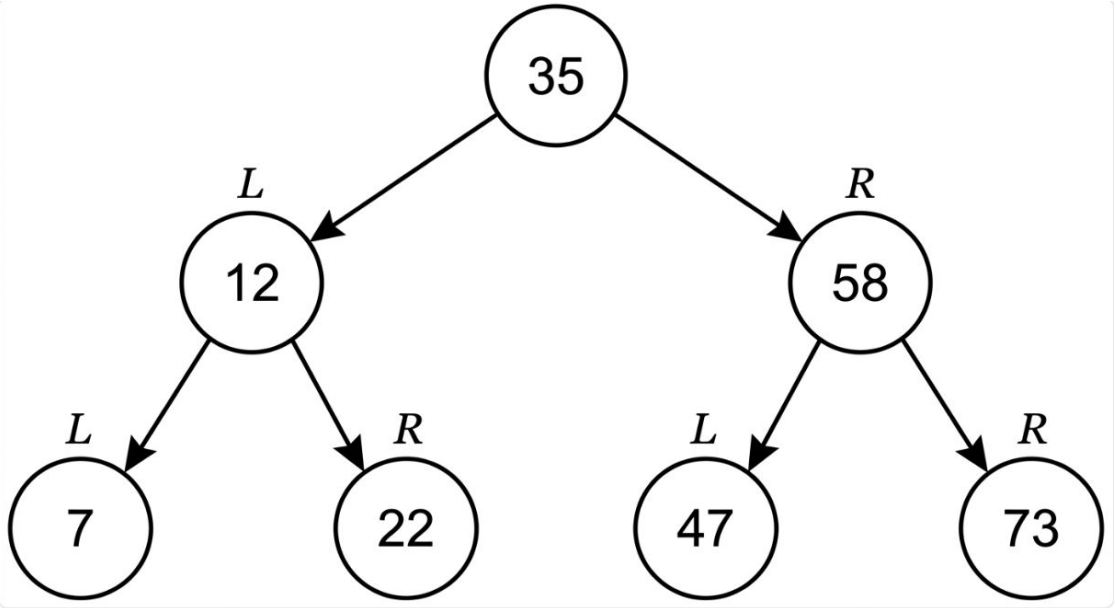
Use Array: [35, 12, 58, 7, 22, 47, 73]

Special instructions / teaching style (optional)

Label left and right children clearly with L and R

☑ Generate alt text ☐ Advanced mode (review scene description)

Generate visualization



Note: A **Binary Search Tree** arranges values so every node's (circle), left children are smaller and right children are larger. This structure lets you find any value quickly by cutting the search in half at each step.

Customizable

Two-step AI pipeline

User input

Concept + level + example data

→

Text AI

Gemini 2.5 Pro outputs structured scene data every element, position, and label.

→

Image AI

Nano Banana draws the diagram from the structured plan.

→

Output

Clean & accurate Images

Why split it up?

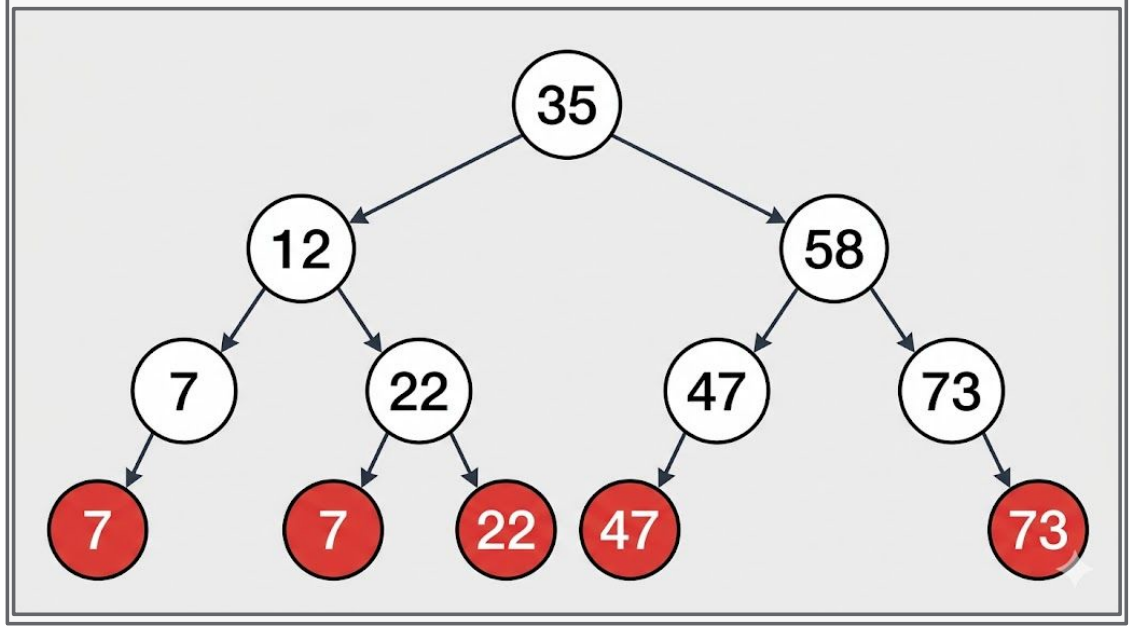
Image models hallucinate. Reasoning models are precise. So the reasoning model outputs a structured JSON plan, exact counts, exact labels, and the image model just renders it. **Nothing is guessed.**

With vs. Without Pipeline

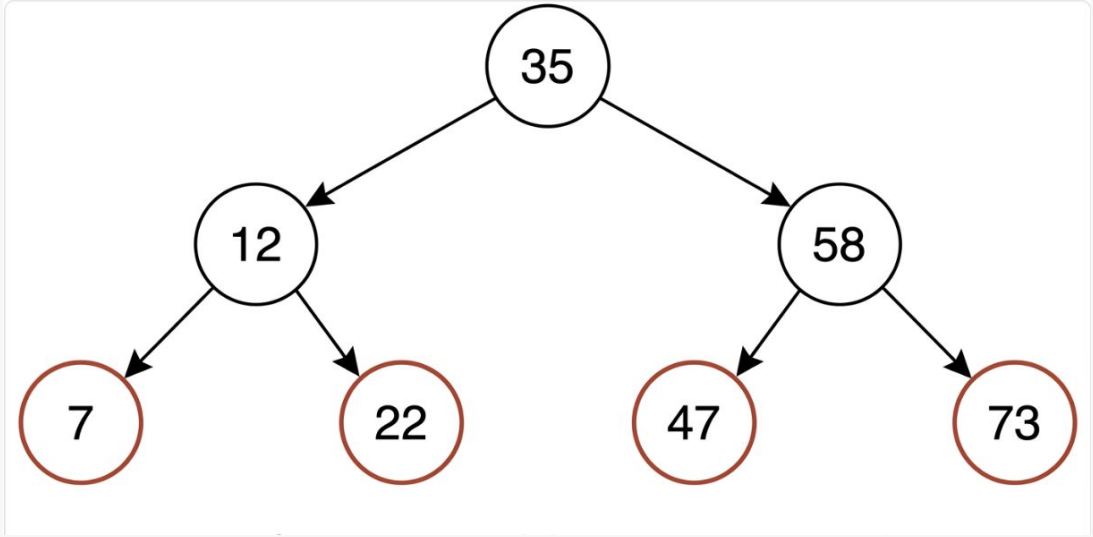
Prompt: Draw a binary search tree containing 35, 12, 58, 7, 22, 47, 73. Highlight the leaf nodes in red.

Note: A leaf is a node with no children – it's at the end of a branch, so 7, 22, 47, and 73 in this case.

WITHOUT PIPELINE



WITH PIPELINE



KEY FEATURE

Step-by-step breakdown

Concept in

"BST insertion" + level + special instructions

→

Classify

AI identifies concept type and picks frame count

→

Step out

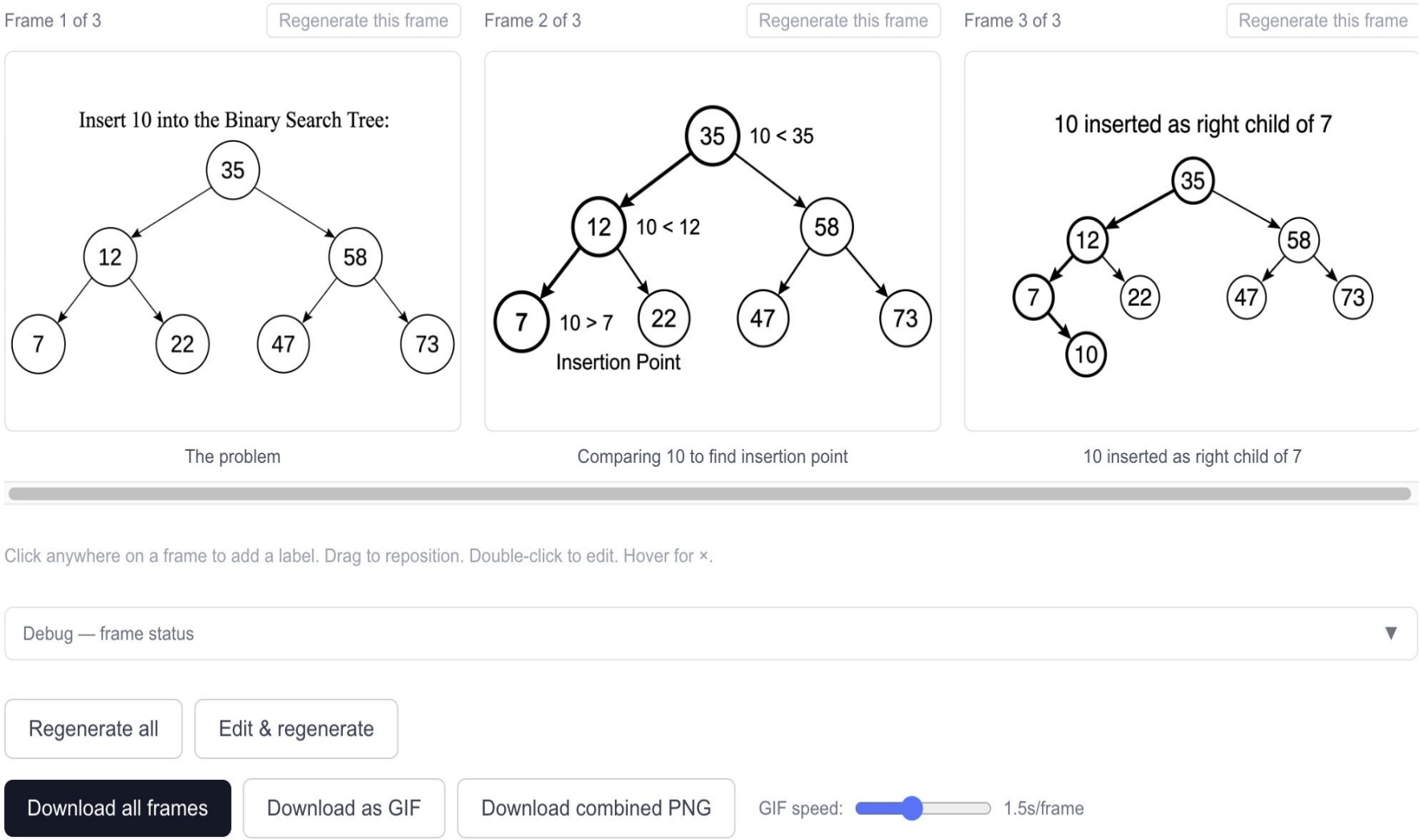
1 image for structures, 2–8 for algorithms

How it decides step count

Static structure → 1 steps. Single operation → 2–3 steps. Multi-step algorithm → 4–6 steps. Full walkthrough → 6–8 steps. **More complicated ideas have more steps so more images.**

Example:

CS concept to visualize:
Binary Tree Insertion
Insert **10** into array
[35, 12, 58, 7, 22, 47, 73.]



KEY FEATURE

Built for every classroom

01

Color mode toggle

Black-and-white by default

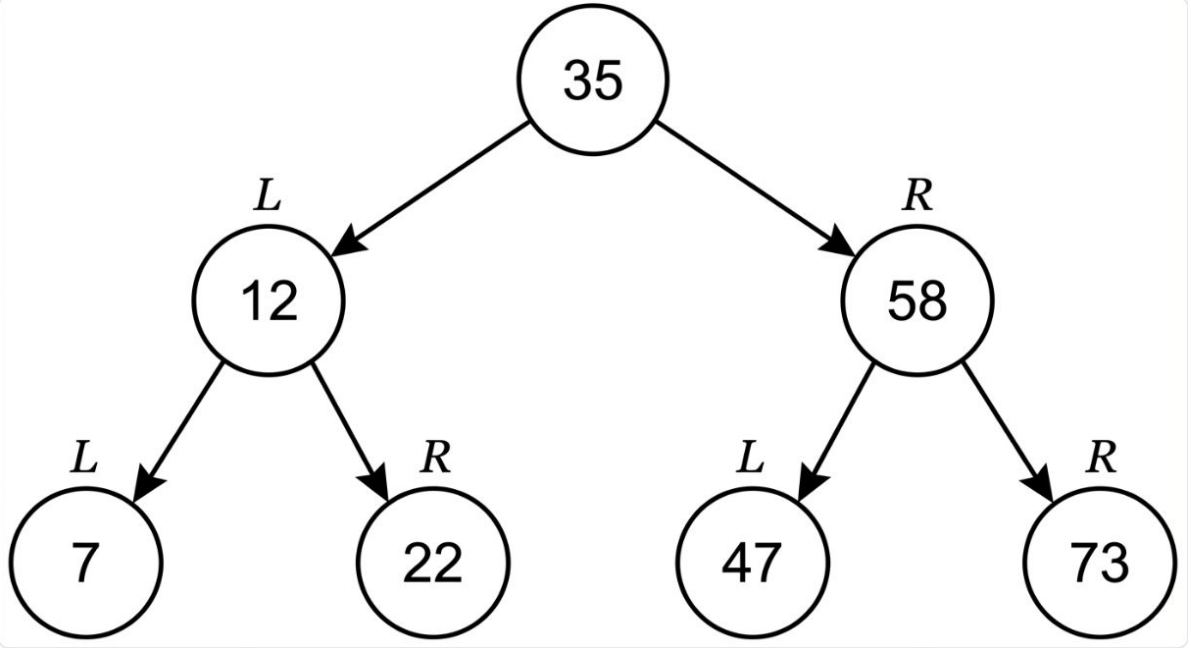
About 1 in 12 people have color vision deficiency. VisualCS defaults to black-and-white line art, so no information depends on color. Switch to color only when needed or prompted.

02

Screen-reader alt text

One click for every image

Every generated visual can produce a screen-reader description, element counts, values, and relationships. It's editable before sharing.



Alt text: A binary tree diagram with 7 nodes. The root node is 35. Its left child is 12 and right child is 58. Node 12 has a left child 7 and right child 22. Node 58 has a left child 47 and right child 73.

Thank you!

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